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(54) Title: ECO-FRIENDLY METHOD FOR CATALYTIC AERIAL OXIDATION OF ALOE-EMODIN TO RHEINAL

(57) Abstract: A process for preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone comprising of aerial oxydation of 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone at oxygen pressure of about 1 to 2 bar in presence of a transition metal complex, a co-catalyst and optionally an oxidizing agent in an organic solvent such as 1,4 dioxane, water or mixtures thereof at a temperature ranging between 25 to 100°C, isolation by appropriate process and optional purification of the product.



WO 2011/030350 A1

**ECO-FRIENDLY METHOD FOR CATALYTIC AERIAL
OXIDATION OF ALOE-EMODIN TO RHEINAL**

Field of Invention:

The present invention describes an environment friendly process for the synthesis of 4,5-dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (rheinal) from 1,8-dihydroxy-3-hydroxymethyl-anthraquinone (aloe emodin) via aerial oxidation in presence of organometallic catalysts. Such organometallic catalysts are selected from oxovanadium (IV), copper (II) and, enzyme containing copper (II).

Background of the Invention:

4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde (compound I), has a diversified synthetic utility in synthesis of various pharmacologically active anthraquinone derivatives. e.g. US5,480,873 describes various anthraquinone derivatives useful for modifying cell functions and are indicated for use in the treatment of skeletal diseases, diabetes and related complications. Most of these anthraquinone derivatives are routed through compound of formula I.

Besides, compound of formula I would have a potential utility in synthesis of diacerhein (compound II, described in French patent No. 2508798), an important therapeutic candidate useful in treatment of degenerative diseases of joints, such as osteoarthritis and connective tissue matrix diseases, such as osteoporosis and rheumatoid arthritis.

A key intermediate of diacerhein viz. 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carboxylic acid (compound III) can be prepared from compound of formula 'I' through well documented oxidation reactions employing heptavalent Mn such as potassium permanganate (CN 101104583), sodium hypochlorite (J. Org. Chem. 71 (2006) 9291-9296), sodium perborate (Tetrahedron 45 (1989) 3299-3306). The stated approach would be an environmentally benign approach when

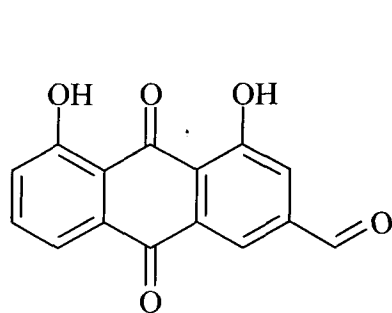
compared to the one reported in WO A-98 56750, involving chromium based catalyst. Hexavalent chromium compounds such as chromic anhydride and dichromate are known for high toxicity and carcinogenicity as well as harmful effect on environment.

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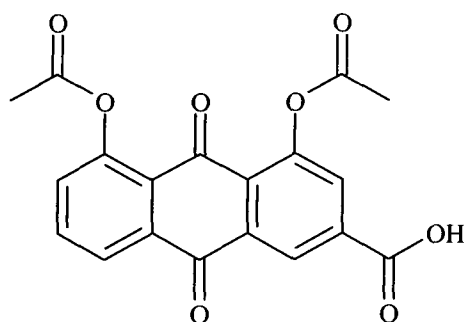
Practice of chromium based process for preparation of diacerein, would require a very cumbersome and tedious process for purification to remove the chromium traces (which is mandatory as per the FDA/ICH guidelines) as described in EP-A-0-636 602, WO-A-00/68179, WO-A-98/56750, WO-A-01/96276, WO-A-2004/050601.

10 The said purification processes are complex, multi-step and involve use of toxic solvents & reagent, and result in decreased yields of diacerein.

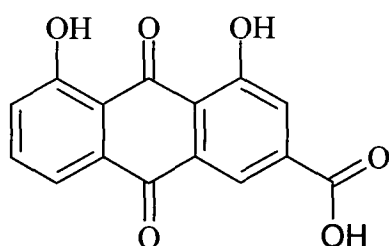
To overcome the described drawbacks related to chromium based process EP-1666446, reports chromium free oxidation using sodium nitrite and 12-14 volume of
15 sulfuric acid to arrive at the compound of formula III. Hence in large scale synthesis it could create difficulty in effluent handling.



(I)



(II)



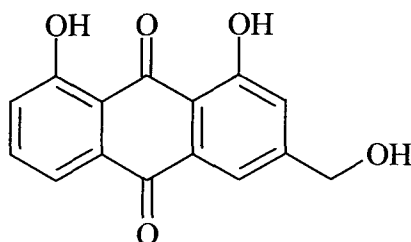
(III)

In view of the facts discussed above there is a need for development of economical, eco-friendly and hazard-free process for preparation of compound of formula I, which in turn would lead to an economical, eco-friendly and hazard-free process for diacerhein and other pharmacologically active anthraquinone derivatives.

Various synthetic approaches has been reported in the literature to overcome some of the problems related to the process for preparation of compound of formula I as provided below.

US 5,480,873 describes a process for preparation of 'I' through modified Swern oxidation of 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone also known as aloe emodin (Compound IV) by employing sulphur trioxide-pyridine complex in dry dimethyl sulfoxide. The by-products of above reaction are dimethyl sulfide, carbon

monoxide, carbon dioxide and when alkyl amine is used as base gives alkylammonium chloride as byproduct. Byproducts generated during the Swern oxidation are dimethyl sulfide and carbon monoxide both toxic and volatile. Further, dimethyl sulfide is a highly volatile liquid with an unpleasant odour.



(IV)

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CN 101104583 describes a process for preparation of 'I' by oxidation of compound IV via well documented literature methods such as chromium trioxide-pyridine complex, pyridine chlorochromate, pyridine dichromate, Dess-Martin catalyst etc. It is to be noted that any oxidation with Cr (IV) would produce Cr (III) salts which are not environmentally benign and require special effluent treatment.

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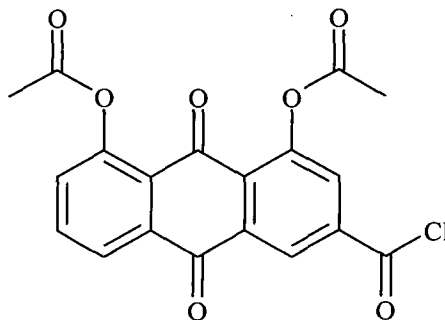
Compound of formula 'I' is also known to be obtained from the extract of *Rheum emodi* as described in Indian Journal of Chemistry, 38 (1999) 749-751. However, isolation using column chromatography, use of large volume of solvents and low yield render the process less attractive for large scale manufacture.

15

Mitter *et al* reported in Journal of the Indian Chemical Society 9 (1932) 375 a synthetic process of compound of formula 'I' involving reduction/hydrogenation of 4,5-diacetoxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbonyl chloride (compound V). Compound V is in turn prepared by chlorination of compound 'II' involving reaction with thionyl chloride. The reduction process involves use of palladium charcoal catalyst poisoned with barium sulfate; otherwise untreated catalyst is too

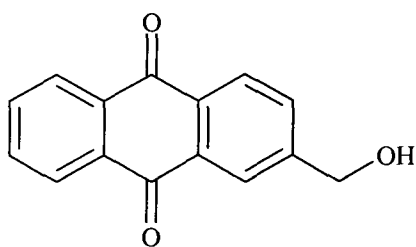
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reactive and gives over reduction products. Hence this could not be a desirable industrial process.

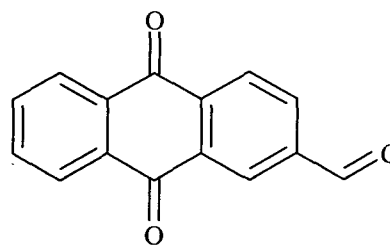


(V)

- 5 Scaiano et al. reported in Chem Commun. (2006) 4401-4403 a photochemical oxidation of 2-(hydroxymethyl) anthraquinone [VI] to 2-formyl- 9, 10-dihydroxyanthracene [VII] Quantum efficiency of this process is low and hence, it could not be utilized for manufacture of compound of formula I.



(VI)



(VII)

- 10 Hence there is a need for development of an eco-friendly *i.e.* "green" and cost effective synthesis of compound 'I'.

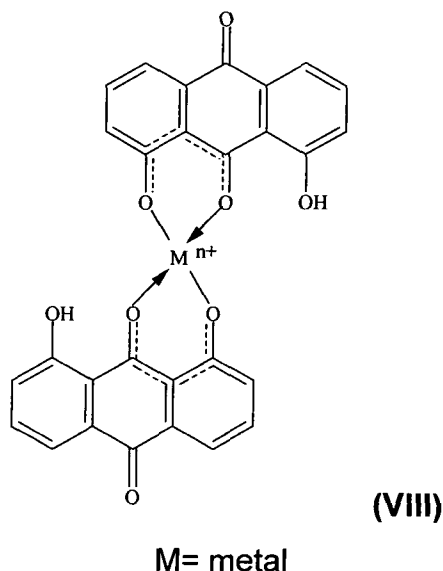
Oxidation of alcohol to corresponding aldehyde in presence of metal catalyst with molecular oxygen is well documented in Tetrahedron 62 (2006) 8227-8241. It is
15 advantageous because oxygen is inexpensive, readily available, and ultimately produces benign byproducts such as water.

Several metals such as cobalt, copper, gold, iron, palladium ruthenium and vanadium as organometallic complex are known for oxidizing alcohol to the respective aldehyde.

- 5 However, several challenges exist in the development of metal-catalyzed aerobic alcohol oxidations. This includes identification of specific metal catalyst, optimization of reaction temperatures, pressures of oxygen and catalyst loading, *etc.* Also nature of metal complex plays a crucial role for oxidation of alcohol. The other key challenge is functional group tolerance and the ability to oxidize an alcohol
10 preferentially in the presence of other functional groups susceptible to oxidation. Hence, there is a need of research for identification of the metal catalysts and optimization of reaction conditions, which would lead towards a cost effective process. This invention provides that.
- 15 Oxidation of protected phenolic alcohol is very well reported with TEMPO (2,2,6,6-Tetramethylpiperidine-1-oxyl)/ Cu(II) catalyzed system as well as organometallic catalysts obtained from Vanadium [Synthesis, (1996) 1153-1174; Tetrahedron Letters, 2006 (47) 922-926; J. Org. Chem. 72 (2007) 7030-7033; Org. Lett. 6 (2004) 217-219; J. Org. Chem. 71(2006) 7087-7090]. It could be rationalized that phenolic
20 group is protected to prevent the oxidation of phenolic 'OH' function, which is likely to give the semiquinone derivative. Since, our system contains the anthraquinone backbone which is much deactivated system. Hence, it was thought that oxidation of phenolic 'OH' function would be less vulnerable and thus could be advantageous because of avoidance of the protection and deprotection of phenolic 'OH', which
25 would be technologically superior as well as economically more feasible over prior art.

However, there are, no report of oxidation of free hydroxy anthraquinone primary alcohol to the corresponding hydroxy anthraquinone aldehyde with TEMPO catalyzes system as well as organometallic catalyst obtained from Vandadium.

- 5 Hydroxy anthraquinones are known to form complex with various simple metal salts, such as copper, cadmium, nickel, iron, vanadium and cobalt called as, "lake" complexes of formula 'VIII'. (Russian Journal of Coordination Chemistry, 2004, (30) 360-364; JACS, 1919 (41) 2081–2083). Hence, it is very challenging to develop a process for oxidation of alcohol function attached to hydroxy anthraquinones. The
10 present invention provides that.



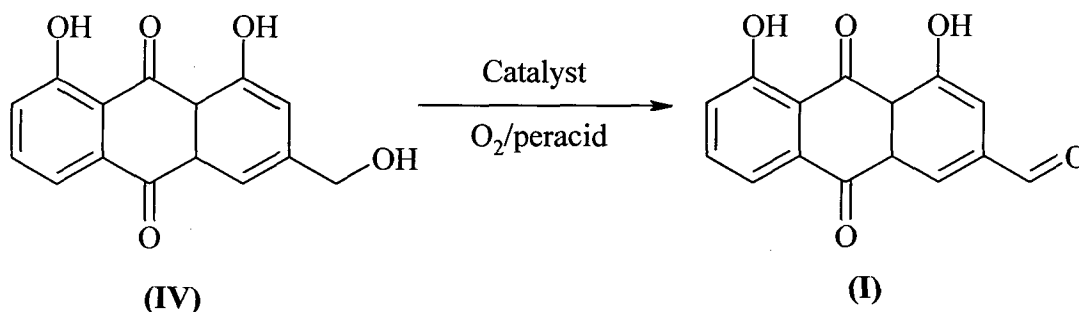
Objects of the invention

- Thus the object of the present invention is to provide cost effective, hazard-free and
15 eco-friendly process for preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde, *i.e.* compound of formula I of high purity.

- A further object of the present invention is to provide an improved and cost effective industrial process for the preparation of compound of formula I that produces
20 minimum by-products.

Summary of Invention:

The present invention provides a process for preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde 'I' by employing aerial oxidation in presence of organometallic catalysts has been described. The said metal catalysts are selected from Oxovanadium (IV)/DABCO (1,4-diazabicyclo[2.2.2]octane), Copper (II)/TEMPO and enzyme containing Copper (II)/TEMPO. To our surprise we found that reaction conducted in presence of a peroxy compound such as *m*-chloroperbenzoic acid (*m*CPBA), the rate of reaction increases approximately two fold and often increases the overall yield.



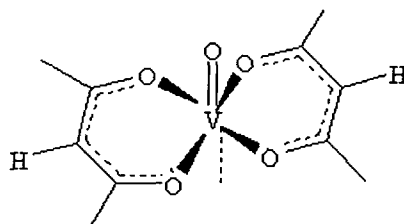
Scheme 1. Reaction Scheme

Detailed Description

The invention embodies transition metal complex catalyzed oxidation of benzylic hydroxyl group of compound of formula IV to corresponding aldehyde of formula I, having purity of 98%.

According to one embodiment of the invention, oxidation of compound 'IV' to corresponding aldehyde 'I' was performed in presence of organometallic catalysts obtained from Oxovanadium (IV), Copper (II), Cobalt (II) and Iron (III); preferably with Oxovanadium (IV) and Copper (II).

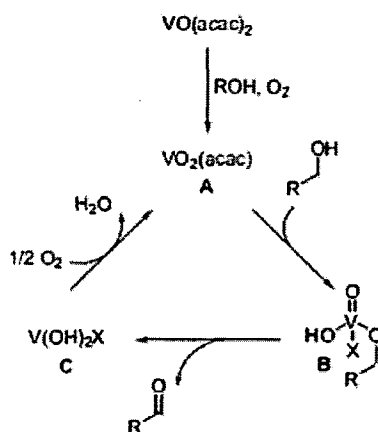
It was observed that rate of reaction and conversion is more with oxovanadium such as $\text{VO}(\text{acac})_2$ (Vanadyl acetylacetonate, compound of formula IX) as compared to other metal complexes such as copper, cobalt and iron.



(IX)

It is well documented that $\text{VO}(\text{acac})_2/\text{DABCO}/\text{O}_2$ oxidize alcohol to the corresponding aldehyde (Tetrahedron 62 (2006) 8227–8241). However, it is to be noted that there are no report of oxidation of hydroxy anthraquinone alcohol to the corresponding aldehyde using such catalyst.

The mechanism for $\text{VO}(\text{acac})_2$ catalyzed oxidation has been rationalized as initial oxidation of V(IV) to V(V) by O_2 followed by the attack of alcohol to form a V-alkoxide. Elimination of the alkoxide would result in product formation and V(III), which could then be reoxidized by O_2 to reform the active V(V). *m*CPBA helps in re-oxidation of V(III). It is depicted in the Scheme 2.

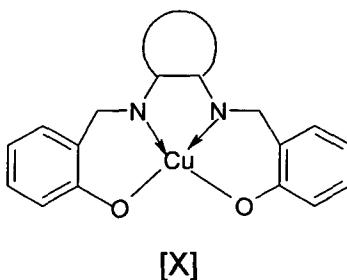


Scheme2 Reaction mechanism of $\text{VO}(\text{acac})_2$ catalyzed oxidation

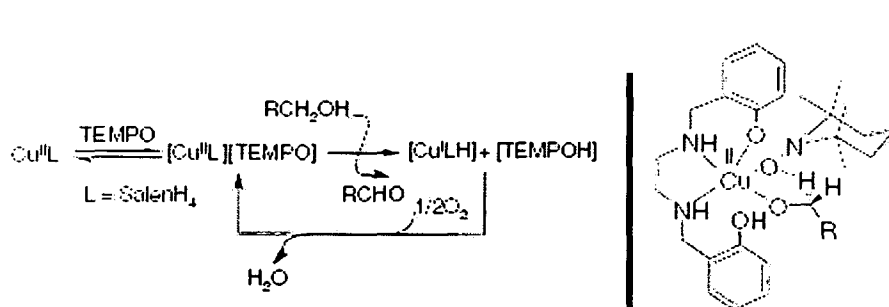
There was no significant improvement in the rate as well as conversion of compound 'IV' to compound 'I' by increasing the oxygen pressure. However, it was observed that addition of *m*CPBA in the VO(acac)₂ /DABCO/O₂ catalyzed reaction increases the rate of reaction of compound [II] to compound [I], which is apparently due to homogenous source of oxygen in the form of *m*CPBA.

TEMPO (tetramethylpiperidinyloxy radical) and its derivatives is versatile oxidant, large number of reports are documented for oxidation of alcohols to corresponding aldehyde are well documented (Synthesis (1996) 1153-1174). For example, TEMPO with aqueous sodium hypochlorite (Adv. Synth. Catal. 346 (2004) 1051-1071); TEMPO with stoichiometric iodine (Organic Letters 5 (2003) 285-287); TEMPO with simple Cu (II) salts and Cu (II) complexes (Tetrahedron Letters, 2006 (47) 922-926) in presence of O₂. However, it is to be noted that there are, no report of oxidation of hydroxy anthraquinone alcohol to the corresponding aldehyde in presence of TEMPO catalyzes systems.

It has to be emphasized that TEMPO catalyzed oxidation in presence of simple salts does not oxidize the hydroxy anthraquinone alcohol to corresponding aldehyde. But TEMPO along with Cu(II) complex such as Cu (II) salen H₄ complex (compound of formula X) oxidizes the hydroxy anthraquinone alcohol (compound IV) to corresponding hydroxy anthraquinone aldehyde (compound I).

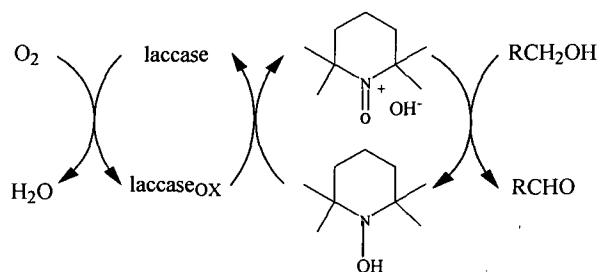
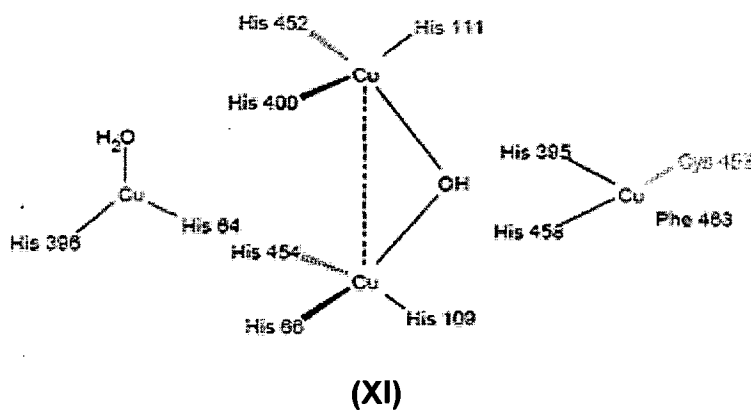


Reaction mechanism for Cu (II) salen H₄ is - it undergoes ligand exchange to form a Cu-alkoxide that binds with TEMPO. The Cu-TEMPO intermediate can proceed to the aldehyde via hydrogen atom abstraction by TEMPO. Molecular oxygen is proposed to reoxidize TEMPOH to TEMPO followed by reoxidation of Cu (I) by TEMPO (Tetrahedron Letters, 2006 (47) 922-926).



Scheme3 Reaction mechanism of Cu (II) salen H₄/TEMPO/O₂ catalyzed oxidation

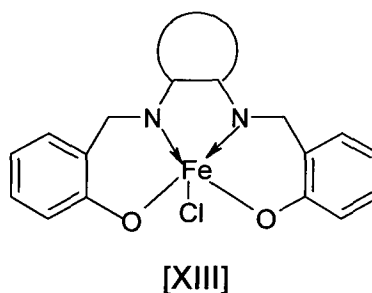
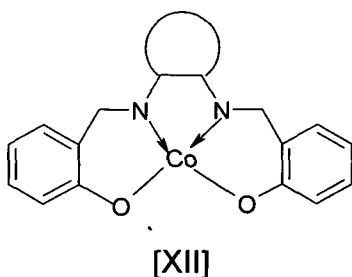
- 10 Addition of 10 mole % of *m*CPBA in the Cu (II) Salen H₄ /TEMPO/ O₂ catalytic system increases the rate of reaction and conversion of compound 'IV' to compound 'I'. This is due to homogenous sources of oxygen in the form of *m*CPBA but there was no increase in rate as well as conversion by increasing the oxygen pressure.
- 15 Laccases (EC 1.10.3.2, p-diphenol:dioxygen oxidoreductase, compound XI) belong to multinuclear copper-containing oxidases. They catalyse the monoelectronic oxidation of substrate at the expense of molecular oxygen (Trends in Biotechnology, 2006 (24) 219-226). Copper (in laccase enzyme) has two N₂ ligands from two histidines and one oxygen ligand (Journal of Biological Chemistry, 2005). However,
- 20 it is to be noted that there are, no report of oxidation of hydroxy anthraquinone alcohol to the corresponding aldehyde.



5 Scheme 4 Reaction mechanism of Laccase/TEMPO/O₂ catalyzed oxidation

The active form of TEMPO is the oxidized product, called oxoammonium ion, which is formed by laccase_{OX}. Laccase_{OX} is generated by oxidation of laccase through oxygen. Oxoammonium ion oxidizes the alcohol to aldehyde liberating back the inactive form of TEMPO (VI) that again gets oxidized. (J. Mol Cat. B: Enzymatic 37 (2005) 79-83).

Although it has been reported that Co(II), Fe(III) salen H₄ complex (compound XII and XIII respectively) along with oxygen oxidizes primary benzylic alcohol to the corresponding aldehyde, but in our system no oxidation product was observed, a fact which is difficult to rationalize.

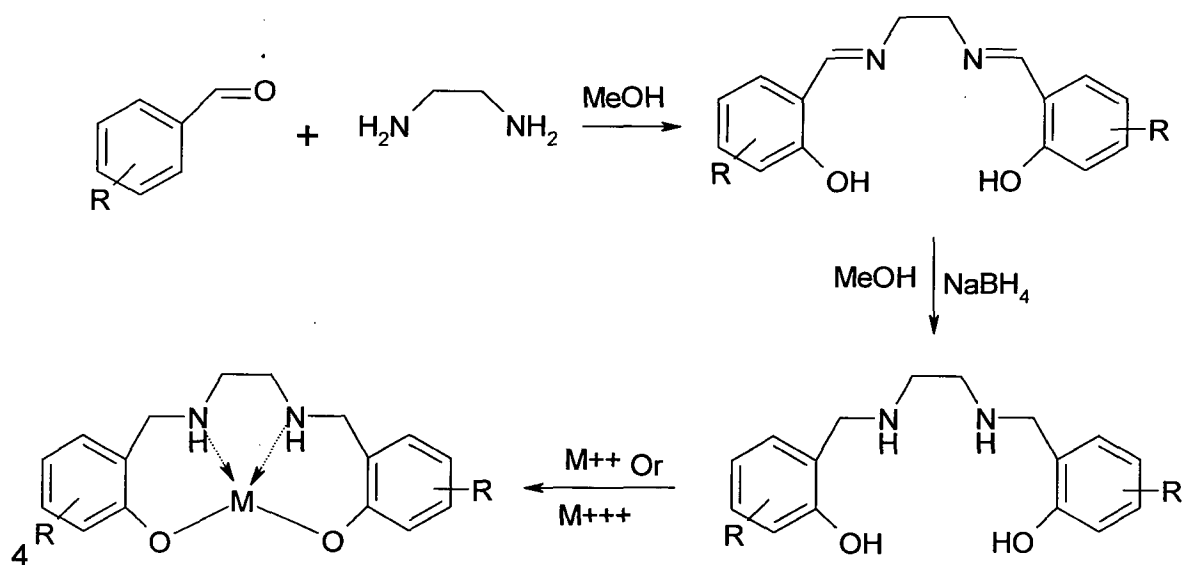


However, Cu (II) salen H₄ complex is an efficient oxidizing catalyst in presence of oxygen/ air more as in presence of a peroxy acid such as *m*-chloroperbenzoic acid, peroxyacetic acid, peroxybenzoic acid, preferably *m*-chloroperbenzoic acid

5

The said salen H₄ complexes were made by the procedure reported in literature. (Tetrahedron Letters, 2006 (47) 922-926), which is schematically shown in **Scheme 5**

5



10

Scheme 5. Synthesis of Metal Salen H₄ complex

The Schiff base is obtained by reaction of an aromatic aldehyde with the diamine.

The aromatic aldehyde is selected from salicylaldehyde or substituted salicylaldehyde; preferably salicylaldehyde.

The diamine component in the Schiff base is aliphatic, substituted aliphatic or cyclic diamine such as ethylene diamine, 1, 2-cyclohexane diamine; preferably ethylene diamine, 1,2-cyclohexyldiamine

Oxovanadium (IV) is in the organometallic complex such as, Oxovanadium (IV) acetylacetonate.

The acetylacetonate is simple acetylacetonate or substituted acetylacetonate.

Compound 'IV' with DABCO and in the presence of Oxovanadium (IV) compounds, such as $\text{VO}(\text{acac})_2$ and molecular oxygen or air as oxidizing agent gives the Compound 'I' in quantitative yield. Wherein, $\text{VO}(\text{acac})_2$ is available from Sigma Aldrich co or could be prepared by literature method (Inorg. Synth. 1957; 5: 113–116).

Compound 'IV' with 1, 4-Diazabicyclo [2.2.2] octane (DABCO) and in the presence of Oxovanadium (IV) compounds, such as $\text{VO}(\text{acac})_2$, *m*-chloroperbenzoic acid and molecular oxygen or air as oxidizing agent gives the Compound 'I' also in quantitative yield but the rate increases 2 fold.

Compound 'IV' with oxaammonium ion, in particular 2,2,6,6 – tetramethyl piperdin-1 oxyl (TEMPO) and in the presence of Cu (II) complexes such as Cu (II) Salen- H_4 [IX] (Tetrahedron Letter, 2006, 47, 923-926) and oxygen gives the Compound 'I'.

The so obtained compound of formula 'I' could be further purified and un-reacted alcohol 'IV' can be recycled to obtain compound 'I'.

Compound 'IV' with oxaammonium ion, in particular 2,2,6,6 – tetramethyl piperdin-1 oxyl (TEMPO) and in the presence Cu (II) containing enzymes such as laccase 'XI' (Trends in Biotechnology, 2006, 24, 219-226) gives the Compound 'I' and compound 'III'. Further purification of mixture of compound 'I' and 'III' is achieved by column chromatography as per condition described in example 4.

Useful enzymes for oxidation of alcohol compound to aldehyde and or carboxylic acid may thus include oxidative enzyme, including laccase. Such enzyme may be obtained from a variety of natural sources, including animal organs and microorganisms. Particularly useful laccase include enzyme derived from the microorganism *Trametes versicolor* such as available from Sigma-Aldrich co.

The oxidation reaction for obtaining compound 'I' with metal catalyst is carried out in solvent such as 1,4 dioxane, dimethyl formamide, dimethyl sulfoxide and N-methyl pyrrolidine; preferably reaction is performed in 1,4- dioxane.

The oxidation reaction for obtaining compound 'I' with metal catalyst is carried out in mixture of solvent such as water and 1,4 dioxane (1:1).

The oxidation reaction for obtaining compound 'I' with laccase enzyme is carried out in buffer having pH range 4-6; preferably at pH 5.

Reaction is generally employ VO(acac)₂ as catalyst, loading of catalyst is about 5% to 20 %; preferably VO(acac)₂ loading of about 5% is used.

Reaction is generally employ Cu (II) salen-H₄ as catalyst, loading of catalyst is about 10% to 30%; preferably Cu (II) salen-H₄ loading of about 10% is used.

Reaction is generally employ laccase enzyme loading of about 10% to about 30%; preferably enzyme loading of about 10% is used

Organometallic catalyzed oxidation is carried out over a wide range of temperature.

- 5 For example, the reaction may be carried out at temperature of about 75 °C to 100 °C, but typically carried out at 90 °C.

Enzymatic reaction is carried out at temperature of 25 to 50 °C, but typically carried out at 25 °C.

- 10 The oxidation is carried out over wide range of oxygen pressure, it was observed that rate of reaction does not have any effect with increasing oxygen pressure hence preferable oxygen pressure is 1 –2 bar.

- 15 Unreacted alcohol 'IV' is recovered from crude product by column chromatography using silica gel as stationary phase and hexane: ethyl acetate (1:1) as mobile phase, which could be recycled, thereby improving the "atom economy" of the overall process.

- 20 The present invention is illustrated in more detail by referring to the following Examples, which are not to be construed as limiting the scope of the invention.

Example 1:

- 25 **Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through VO(acac)₂- DABCO/ O₂ oxidation in 1,4 dioxane as solvent**

- A par autoclave reactor was charged with 200 ml of 1,4 Dioxane to that aloe-emodin (85% purity) (3 g) , VO (acac)₂ (0.150 g), DABCO (0.200 g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and
30 then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was

stirred at 90 °C for 8 h. Reaction was monitored by TLC (1:1, hexane/ethyl acetate). After complete disappearing of starting material (normally after 8hr), reaction mixture was cooled to RT and filtered to remove insoluble material. Filtrate was then poured in 1L of ice cool water to obtain the product. After filtering and washing with water (200 ml), the collected brown solid was dried at 45° C. in vacuo. **Isolated yield:** 2 g (79%); **Melting point:** 202 – 205 °C (Literature melting point 200 °C; USP. 5,480,873); **Compound I:** ¹H NMR (200MHz, DMSO₄ d₆): δ 7.38-8.09 (m, Ar), 10.10(s, Ar-CHO), 11.98-11.95 (m, Ar-OH); **IR Spectra (cm⁻¹):** 1699, 1673, 1630, 1265.

Powder X-ray diffraction:

PXRD Data for 4,5- dihydro 9,10-dioxo 9,10 dihydro anthracene -2-carbaldehyde		
2- Theta	d-spacing [Å]	Rel. Int [%]
8.77	10.07	67.94
8.82	10.01	62.93
9.65	9.16	85.29
11.92	7.42	38.50
14.07	6.29	28.61
24.56	3.62	59.20
25.12	3.31	56.99
27.57	3.23	100
30.61	2.91	10.45

Example2

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through VO(acac)₂- DABCO/*m*CPBA/O₂ oxidation in 1,4 dioxane as solvent

A par autoclave reactor was charged with 200 ml of 1,4 Dioxane to that aloe-emodin (1.5g) (85% purity), VO (acac)₂ (0.075g), DABCO (0.100 g) and *m*CPBA (0.100g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor.

- 5 The resulting mixture was stirred at 90 °C for 5 h. Reaction was monitored by TLC (1:1, hexane/ethyl acetate). After complete disappearing of starting material (normally after 5hr), reaction mixture was cooled to RT and filtered to remove insoluble material.. Workup was done as per given in the example 1. **Isolated yield:** 1.05 g (82%)

10

Example 3

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through VO(acac)₂- DABCO/ O₂ oxidation in 1,4 dioxane: water (1:1) as reaction media

- 15 A par autoclave reactor was charged with 100 ml of 1,4 Dioxane and 100 ml of water to that aloe-emodin (3 g) (85% purity), VO (acac)₂ (0.150 g), DABCO(0.200 g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was stirred at 90 °C for 24 h. Reaction was monitored by TLC (1:1, 20 hexane/ethyl acetate). After complete disappearing of starting material (normally after 8hr), reaction mixture was cooled to RT and filtered to remove insoluble material. Workup was done as per given in the example 1. **Isolated yield:** 1.8gm (70%)

Example-4

- 25 **Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through Cu(II) Salen H₄/TEMPO/O₂ oxidation.**

A par autoclave reactor was charged with 100 ml of 1,4 Dioxane to that aloe-emodin (1 g) (85% purity), Cu(II) Salen H₄ (0.150g), TEMPO (0.150g) were added and

stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was stirred at 95 °C for 8 h and then reaction mixture was cooled to RT and filtered to remove insoluble material. Filtrate was then poured in 1L of ice cool water to obtain the product. After filtering and washing with water (200 ml), the collected brown solid was dried at 45° C. in vacuo. Crude product was further purified by column chromatography using mobile phase hexane: ethyl acetate (1:1) to obtain the pure aldehyde and unreacted compound II could be reused to obtain compound I. **Isolated yield:** 0.2g (24%)

Example-5

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through Cu(II) Salen H₄/TEMPO/O₂/mCPBA oxidation.

A par autoclave reactor was charged with 100 ml of 1,4 Dioxane to that aloe-emodin (1.5 g) (85% purity), Cu(II) Salen-H₄ (0.075g), TEMPO (0.075g), mCPBA (0.100 g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was stirred at 95 °C for 8 h. Workup and purification of crude product was done as per given in example 4 **Isolated yield:** 0.74 (58%)

Example-6

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through Laccase/TEMPO/O₂

A par autoclave reactor was charged with 10 mmole sodium acetate buffer of pH 5 (100ml) to that aloe-emodin (3gm) (85% purity), laccase from *T. versicolor* (0.5gm), TEMPO (0.9 g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in

reactor. The resulting mixture was stirred at 25 °C for 96 h. Reaction was monitored by HPLC gives 34 % carboxylic acid and 48% aldehyde and 15 % un-reacted aloe emodin. Workup and purification of crude product was done as per given in example 4.

5

Chromatographic Condition:

Instrument: HPLC equipped with Pump, Injector, UV detector and Recorder.

Column: Inertsil ODS 3V (4.6 x 150mm), 5µm

Flow rate: 1.0 mL/minute.

10

Detector: UV at 254nm.

Example-7

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through
15 **Cu(II) (OAc)₂/ TEMPO/O₂ oxidation in 1,4 dioxane as a solvent**

A par autoclave reactor was charged with 200ml of 1,4 Dioxane to that aloe-emodin (2gm) (85% purity), Cu(II) (OAc)₂ (0.075gm), TEMPO (0.56g) were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was
20 stirred at 90 °C for 8 h. Reaction was monitored by TLC (1:1, hexane/ethylacetate) and no oxidative product was observed.

Example-8

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-
25 **carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through**
TEMPO/I₂ oxidation in 1,4 dioxane as a solvent

A reactor was charged with 100ml of 1,4 Dioxane to that aloe-emodin (3gm) (85% purity) was added and in that aqueous solution of sodium bicarbonate (1 gm in 10ml of deionized water) added. Solid iodine (5.78gm) was added to above reaction

mixture in one portion followed by TEMPO (0.150gm) and stirred at room temperature for 15 min. Reaction was monitored by TLC (1:1, hexane/ethylacetate) and no oxidative product was observed.

5 **Example-9**

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through Fe (III) salen H₄ / O₂ oxidation in 1,4 dioxane as a solvent

10 A par autoclave reactor was charged with 200ml of 1,4 Dioxane to that aloe-emodin (3gm) (85% purity), Fe (III) salen (0.150gm), were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was stirred at 90 °C for 8 h. Reaction was monitored by TLC (1:1, hexane/ethylacetate) and no oxidative product was observed.

15 **Example-10**

Process of preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone through Co (II) salen H₄ / O₂ oxidation in 1,4 dioxane as a solvent

20 A par autoclave reactor was charged with 200ml of 1,4 Dioxane to that aloe-emodin (2gm) (85% purity), Co (II) salen (0.120gm), were added and stirred at room temperature for 15 min. Reactor was purged with oxygen gas for 5 min and then 1 bar oxygen pressure was maintained in reactor. The resulting mixture was stirred at 90 °C for 8 h. Reaction was monitored by TLC (1:1, hexane/ethylacetate) and no
25 oxidative product was observed.

CLAIMS

- 5 1. A process for preparation of 4,5-Dihydroxy-9,10-dioxo-9,10-dihydro-anthracene-2-carbaldehyde from 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone comprising of aerial oxydation of 1,8-Dihydroxy-3-hydroxymethyl-anthraquinone at oxygen pressure of about 1 to 2 bar in presence of a transition metal complex, a co-catalyst and optionally an
10 oxidizing agent in an organic solvent such as 1,4 dioxane, water or mixtures thereof at a temperature ranging between 25 to 100°C, isolation by appropriate process and optional purification of the product.
2. The process as claimed in claim 1, wherein a transition metal complex is Vanadyl acetylacetonate and a co-catalyst is 1,4-diazabicyclo[2.2.2]octane.
- 15 3. The process as claimed in claim 2 wherein loading of Vanadyl acetylacetonate catalyst is about 5% to 20 % of substrate.
4. The process as claimed in claim 1, wherein a transition metal complex is Copper (II) salen H₄ complex and a co-catalyst is 2,2,6,6-Tetramethylpiperidine-1-oxyl.
- 20 5. The process as claimed in claim 4 wherein loading of catalyst is about 10% to 30% of substrate.
6. The process as claimed in claim 1, wherein a transition metal complex is laccase enzyme and a co-catalyst is 2,2,6,6-Tetramethylpiperidine-1-oxyl.
7. The process as claimed in claim 6 wherein loading of laccase enzyme loading
25 is about 10% to about 30% of substrate.
8. The process as claimed in claim 1 wherein the oxygen pressure maintained between 1 and 2 bar.

9. The process as claimed in claim 1 wherein an oxidizing agent is selected from peroxyacids such as metachloroperbenzoic, perbenzoic acid and peraceticacid.
- 5 10. The process as claimed in claim 2 wherein an oxidizing agent is selected from peroxyacids such as metachloroperbenzoic acid, perbenzoic acid and peraceticacid.
11. The process as claimed in claim 4 wherein an oxidizing agent is selected from peroxyacids such as metachloroperbenzoic acid, perbenzoic acid and peraceticacid.
- 10 12. The process as claimed in claim 6 wherein an oxidizing agent is selected from peroxyacids such as metachloroperbenzoic acid, perbenzoic acid and peraceticacid.

INTERNATIONAL SEARCH REPORT

International application No
PCT/IN2010/000562

A. CLASSIFICATION OF SUBJECT MATTER
INV. C07C46/08 C12P7/66
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
C07C C12P

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, CHEM ABS Data, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

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X	DATABASE CA [Online] CHEMICAL ABSTRACTS SERVICE, COLUMBUS, OHIO, US; ZHANG, YUE ET AL: "Preparation of rhein from aloe-emodin in nonmetallic oxidate medium", XP002620015, retrieved from STN Database accession no. 2009:388992 * abstract & DALIAN GONGYE DAXUE XUEBAO , 28(1), 13-16 CODEN: DGDXP; ISSN: 1674-1404, 2009, -----	1,9
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Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"E" earlier document but published on or after the international filing date
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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.
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Date of the actual completion of the international search

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INTERNATIONAL SEARCH REPORT

International application No

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

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Information on patent family members

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